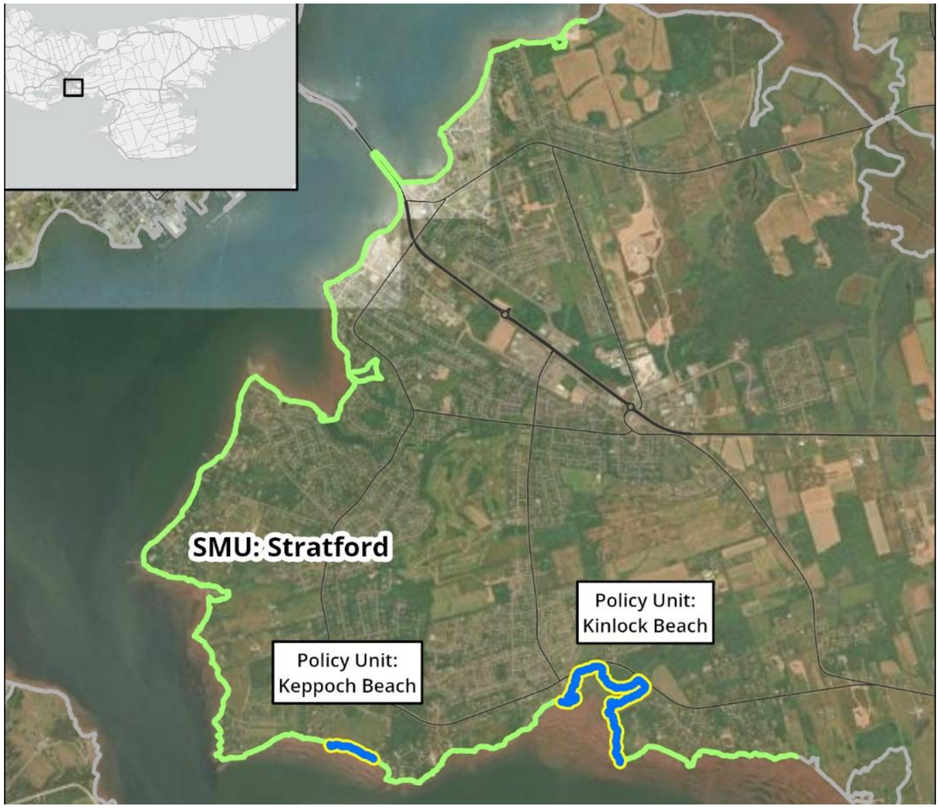
16. Stratford

Shoreline length: 20 km



Shore Types

Stratford is located on the southeast end of PEI. Shore type characterized by a majority of sloped armour, low plain, and cliffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
17	11	39	19	14	0	0	0

Boardwalk and Hillsborough River bridge into Stratford

Stewart Cove

Properties with shoreline armouring along Rosebank

Cove Road point, southeast Stratford

Keppoch Beach

Governance, Infrastructure, and Economy

This area is almost entirely located within the Town of Stratford (96%), where there is predominantly large single-unit dwelling lots along the coastline. Key infrastructure in the coastal area includes the Hillsborough River Bridge (part of the Trans-Canada Highway), the boardwalk, Kinlock Beach, along with Keppoch Road at Kinlock Creek. Stratford is also home to many parks including Robert Cotton Park, Pondside Park, and Kinlock Creek Park. The primary use of the coastline in Stratford is residential, with a small cluster of commercial uses at the connection from the Hillsborough Bridge to the Town. Legacy constraints include densely developed lots located on unpaved (private) roads.

Natural Assets and Heritage

There are a number of parks in the Town that include passive greenspace, walking trails, playgrounds, and tennis and basketball courts: Dr. Leo Frank Park (playground), Partridge Park (playground), Pondside Park (playground, tennis, basketball, hockey, water), Lantz Park (playground), Regent Park (playground, trail), Robert L. Cotton Memorial Park (passive greenspace, trails), and Starling Park (greenspace) as well as Kinlock Beach. Camp Gencheff Rotary Children's Camp and the Stratford Community Garden are also found adjacent to the coastal area. Heritage sites include the Hazard Point Range Front Light.

The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Stratford has municipal planning authority. Current planning documents apply watercourse and wetland regulations in all zones, including an O2 Reserve Zone — not shown on the zoning map — whose boundaries are interpreted to include any area defined as a Wetland or Watercourse, plus the area within 60 metres (197 feet) of a Wetland or Watercourse. Draft documents currently out for consultation would add vertical elevation requirements for habitable floors. Provincial rules apply in the small, unincorporated portion of this SMU. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is low.

Coastal Flood and Erosion Risk

Stratford has a high density of infrastructure close to the shoreline. In some areas the historical armouring does not allow a reliable estimate of background erosion rates, which may be lower as a result. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	2 [<1/km]	6 [<1/km]	19 [1/km]	38 [2/km]	2 [<1/km]	10 [1/km]	40 [2/km]
Roads	163 m [8 m/km]	239 m [12 m/km]	429 m [22 m/km]	942 m [48 m/km]	0	56 m [3 m/km]	135 m [7 m/km]

Proposed SMP Strategies

The primary Resist strategy can be maintained where necessary until increasing flood risk warrants MR in defined areas as follows.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
16.00 Stratford	**R**	**R**	**R**
Rationale	The prevailing Resist strategy may continue along this developed shore where infrastructure is generally outside the flood zone. Flood mitigation should be considered along localized areas.		
16.01 Keppoch beach -PU	**NAI**	**NAI**	**NAI**
Rationale	There are no built assets at risk, and a strategy of No Active Intervention would preserve the beach.		
16.02 Kinlock Beach -PU	**NAI**	**MR**	**MR**
Rationale	An initial strategy of No Active Intervention would preserve the beach. The parking lot is in the flood zone and would need to be raised and/or retreated back towards the road in the medium to long-term.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.